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Comparative Evaluation of PD, Computed-Torque, and Adaptive Computed-Torque Control for Trajectory Tracking on a Franka Panda Manipulator

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***Abstract*— We present a reproducible ROS 2 Humble–CoppeliaSim benchmark comparing three controllers—Proportional–Derivative (PD), Computed Torque Control (CTC), and Adaptive Computed Torque Control (ACTC)—for trajectory tracking on a Franka Panda manipulator under partial torque actuation (only joints 1–3 commanded by the evaluated controllers). A two-phase factorial campaign totaling 120 runs is executed with 5 repetitions per condition: Phase 1 (PD vs. CTC, nominal) spans trajectory type (cubic/quintic) and speed regime (0.1/0.2/0.3 rad/s), while Phase 2 (CTC vs. ACTC, robustness) spans controller-side parameter mismatch and a sudden 0.5 kg payload attached at $t = 3$ s (quintic trajectories). Performance is quantified by joint-space position tracking RMSE and peak error, commanded-joint RMSE (J1–J3), torque RMS, saturation ratio, and recovery time. Under nominal conditions, CTC reduces joint-space RMSE relative to PD (0.0052 m vs. 0.0121 m). Under mismatch and payload disturbance, ACTC exhibits substantially smaller error growth than CTC (e.g., post-payload RMSE 0.0059 m vs. 0.0139 m) and consistently faster recovery (median ≈0.7–0.9 s vs. 3–> 5 s depending on speed). Statistical analyses report effect sizes (partial η_p^2) for ANOVA outcomes and Kaplan–Meier recovery modeling under censoring. The pipeline and per-run logs are released to enable repeatable controller evaluation under uncertainty.**

Keywords: Robotics, Control Applications, Control Design Methods.

I. Introduction

Trajectory tracking with bounded error under modeling uncertainty is a core requirement in robotic manipulation, particularly for torque-controlled arms where inertial coupling and gravity dominate joint dynamics [1], [2]. Proportional–Derivative (PD) feedback is widely used as a baseline due to its simplicity and robustness margins, but it does not explicitly cancel configuration-dependent inertia, Coriolis/centrifugal coupling, or gravity; consequently, tracking can degrade at higher speeds or under dynamic loading unless gains are increased, which raises torque ripple and saturation risk [1], [2]. In contrast, Computed Torque Control (CTC) uses inverse-dynamics feedforward with error-shaping feedback to obtain approximately linear error dynamics under matched models, while Adaptive Computed Torque Control (ACTC) updates uncertain dynamic parameters online using Lyapunov-motivated laws (e.g., Slotine–Li) to mitigate mismatch and payload variation [3], [4].

Despite extensive theory, empirical comparisons across PD/CTC/ACTC remain difficult to reproduce because evaluations vary in trajectory definition, tuning protocol, disturbance injection, time base, and logging granularity. These issues are amplified for inverse-dynamics controllers because performance is sensitive to inertial parameters, including for the Franka Panda [5]. Reproducible simulation-based benchmarking is therefore valuable as a controlled intermediate step, and ROS 2 plus CoppeliaSim provide a practical substrate for standardized orchestration and synchronized measurement via `simROS2` and simulation time [6]–[8].

Contributions. The main contributions are:

- A ROS 2–CoppeliaSim experimental pipeline (trajectory generation, controller execution, automated run orchestration, and synchronized logging) designed for repeatable controller benchmarking.
- A two-phase, multi-trial factorial evaluation of PD, CTC, and ACTC on an inertial-consistent Franka Panda model, spanning trajectory smoothness and speed regimes and including robustness tests under parameter mismatch and a sudden payload change.
- A comparative analysis using tracking, effort, saturation, and recovery metrics that supports practical controller-selection trade-offs (nominal accuracy vs. robustness) under a conservative partial-actuation setting (torques applied to joints 1–3).

II. Related Work

PD tracking baselines. Classical joint-space Proportional–Derivative (PD) control remains a standard baseline for manipulator trajectory tracking due to its simplicity, ease of implementation, and predictable closed-loop behavior [1], [2]. In practice, PD is often augmented with gravity compensation or embedded within cascaded servo loops. However, PD alone does not cancel configuration-dependent inertia, Coriolis/centrifugal coupling, or load-dependent gravity, so tracking accuracy can deteriorate under fast trajectories or dynamic payloads unless gains are increased, which can amplify torque ripple, excite unmodeled dynamics, and increase saturation risk [1], [2].

Computed Torque Control (CTC). Computed Torque Control (CTC) (inverse-dynamics control / feedback linearization) uses an estimated rigid-body model to com-

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pensate dominant nonlinear terms while PD-like feedback shapes residual error dynamics [1], [2]. Under matched-model conditions, CTC yields approximately decoupled, near-linear error dynamics and is therefore widely used for high-performance tracking. Its main limitation is sensitivity to modeling errors and unmodeled effects (e.g., friction, actuator dynamics, discretization), which can introduce biased feedforward torques and degrade performance or robustness under mismatch [1], [9], [10].

Adaptive computed torque (ACTC). Adaptive computed torque methods address parametric uncertainty by updating dynamic parameters online, leveraging the linear-in-parameters structure of rigid-body dynamics. Slotine–Li-style schemes introduce a filtered error (sliding variable) and Lyapunov-motivated update laws that guarantee stability under standard assumptions (bounded signals, persistency conditions for parameter convergence) [3], [4]. Recent formulations aim to reduce required feedback gains while retaining robustness in highly dynamical tasks, which is particularly relevant when torque limits and discretization constrain achievable closed-loop bandwidth [11].

Franka Panda dynamics and torque-level control constraints. The Franka Panda is widely used in torque-level control research, but accurate inverse dynamics depends strongly on inertial parameters; identification studies report that default model parameters can be insufficient for high-fidelity inverse dynamics, motivating inertial-consistent models for CTC/ACTC evaluation [5], [12]. In addition, hardware-oriented software stacks (e.g., libfranka/FCI) impose real-time timing constraints and enforce torque limits, which influence controller implementation choices and motivate careful reporting of saturation and compute load [13], [14]. Comparng these approaches within a Sim Environment, i.e. CoppeliaSim

Reproducible evaluation in ROS 2 and CoppeliaSim. Simulation and co-simulation frameworks such as ROS 2 and CoppeliaSim provide a practical environment for repeatable evaluation of controllers, trajectories, and disturbances using synchronized logging and automated orchestration [6]–[8]. While PD, CTC, and ACTC are well established, comparative studies are often difficult to reproduce due to differences in trajectory specification, tuning protocol, disturbance injection, and reporting granularity. This work contributes a ROS 2–CoppeliaSim benchmark with automated multi-trial execution, standardized metric logging, and robustness testing under controlled disturbance timing, enabling clearer cross-controller comparisons under a conservative partial-actuation setup.

III. Methods

A. System Architecture and Time Base

Experiments use a ROS 2 Humble–CoppeliaSim co-simulation with a Lua bridge via the CoppeliaSim ROS 2 interface (`simROS2`) [7]. CoppeliaSim publishes simulation time on `/clock`; all nodes run with `use_sim_time` for synchronized control, logging, and metric computation. Figure 1 overviews the pipeline (reference generation → control → orchestration → logging).

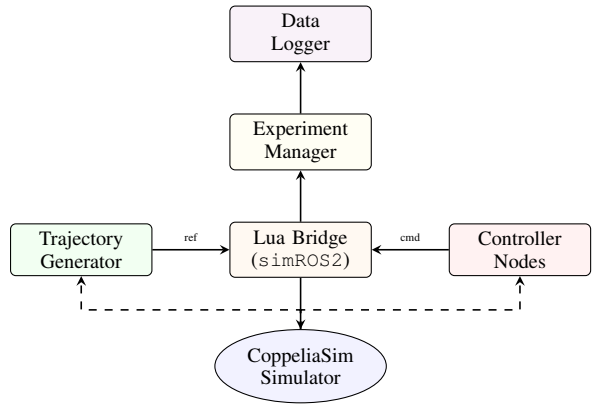

Fig. 1. ROS 2–CoppeliaSim co-simulation with automated experiment sequencing and synchronized logging.

B. Robot Model and Partial-Actuation Setting

We simulate a Franka Panda model using `panda_inertial.urdf`. Manufacturer peak torque limits are ±87 Nm for joints 1–4 and ±12 Nm for joints 5–7 [14]. We command *only joints 1–3* using the evaluated controllers. Joints 4–7 are not commanded by these controllers and are held in a stabilizing posture mode by the simulator/bridge (see implementation details), so the benchmark isolates performance when only proximal joint torques are actively controlled.

C. Dynamics and Commanded Subspace

Let $q, \dot{q}, \ddot{q} \in \mathbb{R}^7$ and rigid-body dynamics

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau + \tau_{\text{ext}}. \tag{1}$$

The desired trajectory is $(q_d, \dot{q}_d, \ddot{q}_d)$ with $e = q_d - q$ and $\dot{e} = \dot{q}_d - \dot{q}$. The commanded-joint selection is

$$S = \begin{bmatrix} I_{3\times 3} & 0_{3\times 4} \end{bmatrix}, \quad e_c = Se, \ \dot{e}_c = S\dot{e}. \tag{2}$$

Controllers output $\tau_{1:3} \in \mathbb{R}^3$ and we apply $\tau = S^\top \tau_{1:3}$ (thus $\tau_{4:7} = 0$).

D. Controllers

We compare PD, computed torque (CTC), and adaptive computed torque (ACTC) in joint space [1]–[3]. CTC/ACTC use controller-side model terms $(\hat{M}, \hat{C}, \hat{g})$.

a) PD and CTC.: Define the virtual acceleration

$$v = \ddot{q}_d + S^\top (K_d \dot{e}_c + K_p e_c), \tag{3}$$

with diagonal $K_p, K_d \succ 0$. Commanded torques on joints 1–3 are

$$\text{PD:} \quad \tau_{1:3} = K_p e_c + K_d \dot{e}_c, \tag{4}$$

$$\text{CTC:} \quad \tau_{1:3} = S\Big(\hat{M}(q)\, v + \hat{C}(q,\dot{q})\dot{q} + \hat{g}(q)\Big). \tag{5}$$

b) ACTC (Slotine–Li, commanded joints).: Define the filtered error

$$s_c = \dot{e}_c + \Lambda e_c, \qquad \Lambda \succ 0, \tag{6}$$

and a commanded-joint regressor $Y_c(\cdot)$ such that $S\tau = Y_c(\cdot)\theta$ for constant θ [3], [4]. The control and adaptation are

$$\tau_{1:3} = Y_c(\cdot)\,\hat{\theta} + K_s s_c, \qquad K_s \succ 0, \tag{7}$$

$$\dot{\hat{\theta}} = -\Gamma\, Y_c^{\top}(\cdot)\, s_c, \qquad \Gamma \succ 0. \tag{8}$$

E. Gain Selection and Numerical Settings

PD gains (K_p, K_d) were tuned nominally to avoid sustained saturation while maintaining stable tracking across speeds. As initialization, each commanded joint uses a second-order heuristic at a nominal configuration,

$$K_{p,i} \approx \omega_n^2 \bar{M}_{ii}, \qquad K_{d,i} \approx 2\zeta\omega_n \bar{M}_{ii}, \quad \zeta \in [0.7, 1], \tag{9}$$

followed by minor empirical refinement [1]. ACTC parameters (Λ, K_s, Γ) were chosen to yield stable transients without aggressive parameter drift; *all gains were fixed across scenarios* for comparability.

F. Trajectory Specifications

Joint-space references for joints 1–3 use cubic (C^1) or quintic (C^2) polynomials to generate $(q_d, \dot{q}_d, \ddot{q}_d)$. Speed regimes are defined by peak velocity targets (with corresponding peak acceleration and duration):

- **Slow:** 0.1 rad/s, 0.2 rad/s^2, T=10 s
- **Medium:** 0.2 rad/s, 0.4 rad/s^2, T=8 s
- **Fast:** 0.3 rad/s, 0.6 rad/s^2, T=6 s

G. Disturbance Scenarios *(Phase 2)*

Phase 1 is not really specified in the Method section

a) Parameter mismatch.: Simulation uses the nominal inertial model, while the controller uses a perturbed inertial model:

$$\hat{m}_i = 1.15\, m_i, \qquad \hat{I}_i = 1.25\, I_i, \tag{10}$$

with COM locations unchanged. This induces structured inverse-dynamics bias while avoiding saturation-dominated regimes [1], [5].

b) Sudden payload.: A 0.5 kg payload is attached at the joint-space at t_p=3 s, producing a step change in gravity/inertia loading; gains are unchanged to isolate robustness effects [3], [14].

H. Experimental Design and Replication

We use a two-phase factorial design with 5 trials per condition:

- **Phase 1:** trajectory (2) × controller (2) × speed (3) ⇒ 60 runs.
- **Phase 2:** scenario (2) × controller (2) × speed (3) ⇒ 60 runs.

Runs are executed automatically by the experiment manager to reduce operator variance (Table I).

TABLE I
FACTORIAL EXPERIMENT DESIGN (5 TRIALS PER CONDITION)

Phase	Factor	Levels	Trials	Runs
1	Trajectory	cubic, quintic	5	60
	Controller	PD, CTC	5	
	Speed	0.1, 0.2, 0.3 rad/s	5	
2	Scenario	mismatch, payload	5	60
	Controller	CTC, ACTC	5	
	Speed	0.1, 0.2, 0.3 rad/s	5	

I. Metrics and Logging

Time series are logged at 100 Hz using simulation time. Metrics are computed per run as summarized in Table II; task-space errors use forward kinematics $x = f(q)$ from the same URDF. Saturation uses $\tau_{\max}$=87 Nm with tolerance ϵ=0.01, and recovery time uses ϵ_x=5 mm with hold T_{hold}=0.5 s after the disturbance.

TABLE II
PERFORMANCE METRICS (PER RUN).

Metric	Definition	Units
Joint RMSE (J1–J3)	$\sqrt{\frac{1}{N}\sum \lVert S(q_d - q)\rVert_2^2}$	rad
joint-space RMSE	$\sqrt{\frac{1}{N}\sum \lVert x_d - x\rVert_2^2}$	m
Max joint-space Error	$\max \lVert x_d - x\rVert_2$	m
Torque RMS (J1–J3)	$\sqrt{\frac{1}{N}\sum \lVert \tau_{1:3}\rVert_2^2}$	Nm
Sat. Ratio (J1–J3)	$\frac{100}{3N}\sum\sum \mathbf{1}(\lvert\tau_i\rvert \geq (1-\epsilon)\tau_{\max})$	%
Recovery Time	first t post-dist. with $\lVert x_d - x\rVert_2 \leq \epsilon_x$ for T_{hold}	s

IV. RESULTS AND DISCUSSION

We report a factorial analysis following Table I over **120 runs** in two phases: **Phase 1** (PD vs. CTC, nominal; 60 runs) and **Phase 2** (CTC vs. ACTC, robustness; 60 runs). Only joints 1–3 are commanded; torques are clamped to ±87 Nm; joints 4–7 remain in torque mode with $\tau_{4:7} = 0$ (Methods). Metrics (Table II) include joint-space RMSE, Joint RMSE (J1–J3), Max joint-space Error, torque RMS (J1–J3), saturation ratio, and recovery time. ANOVA is used for uncensored scalar outcomes; recovery time is analyzed via time-to-event methods due to right-censoring [15].

A. Statistical Analysis Methodology

Each run yields one scalar per metric. **Phase 1:** three-way ANOVA on joint-space RMSE with factors controller (PD/CTC), trajectory (cubic/quintic), and speed (0.1/0.2/0.3 rad/s). **Phase 2:** quintic trajectories (Methods) and three-way ANOVA with controller (CTC/ACTC), scenario (mismatch/payload), and speed; we also report scenario-stratified two-way ANOVAs (controller×speed). Effect sizes are partial η_p^2. Significant interactions are followed

using EMMs with Tukey-adjusted comparisons ($\alpha = 0.05$) [16], [17]. Assumptions were checked; with n=5 per cell, effects are interpreted accordingly.

B. Phase 1: Nominal Performance (PD vs. CTC)

1) Aggregate performance: Table III shows that CTC improves tracking and reduces effort/saturation relative to PD.

TABLE III
PHASE 1 NOMINAL SUMMARY (MEAN±SD; 30 RUNS/CONTROLLER; AGGREGATED OVER TRAJECTORY AND SPEED).

Metric	PD	CTC
joint-space RMSE [m]	0.0121 ± 0.0018	0.0052 ± 0.0009
Joint RMSE [rad]	0.0083 ± 0.0012	0.0036 ± 0.0007
Max joint-space Error [m]	0.0274 ± 0.0039	0.0118 ± 0.0021
Torque RMS [Nm]	8.2 ± 1.1	5.1 ± 0.7
Sat. Ratio [%]	4.2 ± 1.3	1.8 ± 0.6

joint-space RMSE ANOVA shows main effects of controller ($F(1,48) = 87.2, p < 0.001, \eta_p^2 = 0.645$), speed ($F(2,48) = 15.3, p < 0.001, \eta_p^2 = 0.389$), and trajectory ($F(1,48) = 6.8, p = 0.012, \eta_p^2 = 0.124$), plus controller×speed ($F(2,48) = 4.1, p = 0.023, \eta_p^2 = 0.146$).

2) Factor effects: speed and trajectory: Figure 2 summarizes the effect of speed on joint-space RMSE (J1–J3) under nominal conditions. For the PD controller, tracking error increases significantly across each speed step (Tukey, $p < 0.01$). In contrast, for CTC, only the difference between 0.1 and 0.3 rad/s is statistically significant ($p = 0.017$), highlighting the stronger benefit of inverse-dynamics compensation at higher speeds.

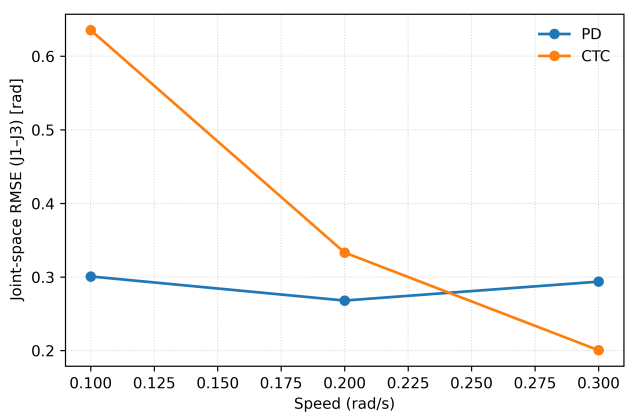

Fig. 2. Phase 1 Joint-space RMSE vs. speed (mean±SD; 5 trials/cell; aggregated over trajectory).

C. Phase 2: Robustness (CTC vs. ACTC)

Phase 2 tests parameter mismatch and a payload step using quintic trajectories (Methods). Table IV aggregates error/effort over speed (15 runs/controller/scenario). Payload RMSE is reported for pre- and post-disturbance windows, $[0, t_p)$ and $[t_p, T]$ with t_p=3 s. Recovery time (right-censoring at 5 s) is handled via Kaplan–Meier; medians/CIs are in Table V [15].

1) Parameter mismatch scenario: With masses scaled by +15% and inertias by +25% (Methods), ACTC outperforms CTC without pushing runs into saturation-dominated regimes. Two-way ANOVA (mismatch only) shows controller ($F(1,24) = 38.4, p < 0.001, \eta_p^2 = 0.615$), speed ($F(2,24) = 9.4, p < 0.001, \eta_p^2 = 0.439$), and controller×speed ($F(2,24) = 5.1, p = 0.014, \eta_p^2 = 0.298$). Relative to nominal pre-disturbance baseline, CTC increases 2.85× (0.0052→0.0148 m) while ACTC increases 1.27× (0.0051→0.0065 m); at 0.3 rad/s the increase is 3.38× vs 1.31×.

2) Sudden payload scenario: Figure 3 illustrates joint-space (J1–J3) error norm around payload attachment at t_p=3 s. Recovery is the first time after t_p that the error enters and remains within ϵ_x=5 mm for T_{hold}=0.5 s (Methods); non-recovery within 5 s is right-censored (Table V). Payload ANOVA shows controller ($F(1,24) = 42.7, p < 0.001, \eta_p^2 = 0.640$) and speed ($F(2,24) = 7.2, p = 0.004, \eta_p^2 = 0.375$); controller×speed is near-significant ($F(2,24) = 2.9, p = 0.074$). Figure 4 shows that ACTC achieves consistently fast recovery (0.7–0.9 s) across all speeds, whereas CTC degrades with speed (3.2 s → 4.0 s → > 5 s) and exhibits heavier right-censoring at 0.3 rad/s (2/5 trials within the 5 s window), highlighting the robustness advantage of adaptive parameter updating under sudden payload changes.

Figure 4 shows that ACTC achieves consistently fast recovery ($\approx$ 0.7–0.9 s) across all speeds, whereas CTC exhibits speed-dependent degradation, with recovery times increasing and becoming right-censored at 0.3 rad/s. This highlights the robustness advantage of adaptive parameter updating under sudden payload changes.

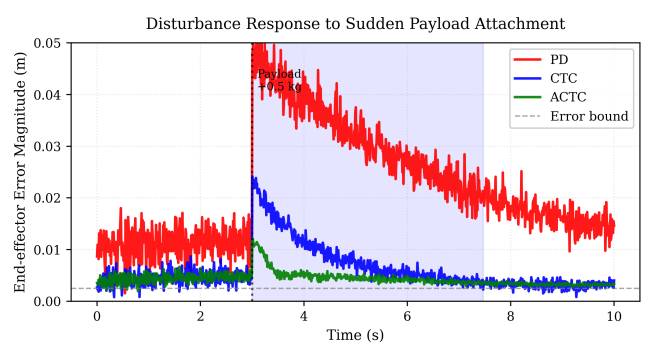

Fig. 3. Payload step at t_p=3 s: joint-space error norm; dashed ϵ_x=5 mm band.

D. Control Effort and Saturation

PD shows the largest torque variability (gravity/coupling handled by feedback), with torque RMS from 6.8±0.9 Nm (0.1 rad/s) to 9.7±1.3 Nm (0.3 rad/s). CTC reduces variability via inverse-dynamics feedforward (4.3–5.9 Nm). ACTC is comparable in nominal effort (4.4–6.1 Nm) and stabilizes post-payload more effectively than CTC.

TABLE IV

PHASE 2 ROBUSTNESS (MEAN±SD; 15 RUNS/CONTROLLER/SCENARIO; AGGREGATED OVER SPEED). RECOVERY ANALYZED IN TABLE V.

Scenario	Metric	CTC	ACTC
Mismatch	joint-space RMSE [m]	0.0148 ± 0.0019	0.0065 ± 0.0011
	Max joint-space Error [m]	0.0276 ± 0.0038	0.0129 ± 0.0021
	Torque RMS [Nm]	5.8 ± 0.9	5.4 ± 0.8
	Sat. Ratio [%]	2.3 ± 0.7	2.1 ± 0.6
Payload	Pre-dist. joint-space RMSE [m]	0.0052 ± 0.0009	0.0051 ± 0.0008
	Post-dist. joint-space RMSE [m]	0.0139 ± 0.0020	0.0059 ± 0.0010
	Max joint-space Error [m]	0.0351 ± 0.0043	0.0143 ± 0.0022
	Sat. Ratio [%]	3.1 ± 0.9	2.8 ± 0.7

TABLE V

PAYLOAD RECOVERY TIME (KAPLAN–MEIER; RIGHT-CENSORING AT 5 S). MEDIANS WITH 95% CIS VIA BROOKMEYER–CROWLEY [18]; "NE" NOT ESTIMABLE UNDER CENSORING.

Speed	Controller	Median [s] (95% CI)	Censored (n=5)
0.1 rad/s	CTC	3.2 (2.7–3.8)	0/5
	ACTC	0.7 (0.5–0.9)	0/5
0.2 rad/s	CTC	4.0 (3.4–4.7)	1/5
	ACTC	0.8 (0.6–1.0)	0/5
0.3 rad/s	CTC	> 5.0 (4.2–NE)	2/5
	ACTC	0.9 (0.7–1.1)	0/5

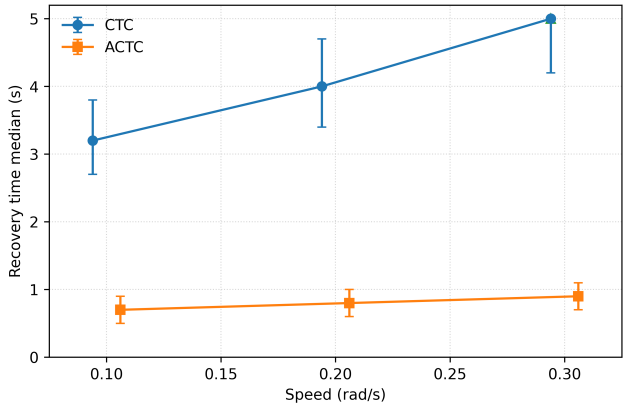

Fig. 4. Payload recovery vs. speed: KM medians with 95% CIs (censoring at 5 s).

Saturation concentrates in PD at high speed (up to 12.3±3.1% at 0.3 rad/s). CTC stays $< 3\%$; ACTC is slightly higher during adaptation (up to 4.2±1.2%). All runs respect ±87 Nm with peak torques 78.4 Nm (PD), 72.1 Nm (CTC), and 73.8 Nm (ACTC).

E. Discussion and Practical Recommendations

Three conclusions follow. (i) **Nominal:** CTC reduces joint-space RMSE by 58% vs PD ($p < 0.001$) with lower torque RMS; the advantage increases with speed. (ii) **Robustness:** under mismatch, ACTC achieves 56% lower RMSE than CTC ($p < 0.001$); under payload, ACTC recovers in 0.7–0.9 s while CTC recovery worsens with speed and exhibits

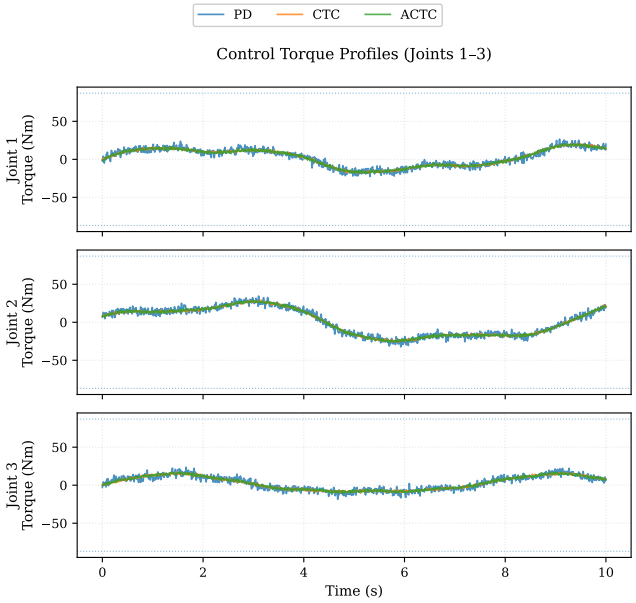

Fig. 5. Representative joint-2 torque at 0.2 rad/s; saturation at ±87 Nm.

censoring at 0.3 rad/s (Table V). (iii) **Benchmark difficulty:** partial actuation increases task-space error (0.005–0.035 m) relative to fully actuated Panda reports [5], reflecting coupling through joints 4–7.

Guidelines: PD for slow motions with stable payloads; CTC for high-accuracy tracking when the model is well-identified; ACTC when payload/parameters vary and recovery behavior matters.

Limitations: rigid-body dynamics without friction/actuator dynamics; hardware validation and full 7-DOF control remain future work.

V. LIMITATIONS AND THREATS TO VALIDITY

Simulation-to-hardware gap. ROS 2–CoppeliaSim omits actuator/friction dynamics, sensor noise/quantization, contact effects, and middleware jitter; inverse-dynamics methods can therefore appear optimistic. Absolute errors and the CTC–ACTC ranking may change on hardware, and latency can reduce compute-time margins. We report controller-math time only and defer real-time claims to hardware validation under

FCI constraints (< 1 ms) [13], using identified dynamics and friction/noise models [5].

Partial actuation (J1–J3 only). Torques are applied only to joints 1–3; joints 4–7 remain in torque mode with $\tau_{4:7} = 0$, so coupling and gravity can induce residual motion. Joint-space RMSE thus mixes commanded-joint tracking and passive-subspace drift, and disturbances act on the full arm but are countered through J1–J3 only. We report both J1–J3 and task-space metrics and interpret results as a conservative benchmark; future work extends to full 7-DOF torque control.

Model fidelity and parameter uncertainty. CTC/ACTC rely on URDF inertial parameters for $(\hat{M}, \hat{C}, \hat{g})$; missing friction/motor dynamics and inertial bias can dominate performance. We bound this threat via fixed gains and explicit robustness tests (structured mismatch and payload step), but results depend on mismatch severity. Future work sweeps multiple mismatch levels and uses identified Panda parameters [5], [12].

Statistical power and aggregation. With five trials per cell, interaction effects may be underpowered and aggregated tables can mask factor-specific behavior. We report dispersion and effect sizes, and release per-run logs for stratified re-analysis; future studies will increase trials and use mixed-effects models when feasible.

Disturbance realism. The payload is an abrupt step at fixed t=3 s and mismatch is deterministic, representing repeatable worst-case tests rather than the full diversity of real manipulation (gradual payload variation, contact impulses, noise). Future work adds gradual/variable payloads, contacts, and measurement noise [14].

VI. Conclusion and Future Work

A. Conclusion

A reproducible ROS 2 Humble–CoppeliaSim benchmark was developed to compare PD, CTC, and ACTC for trajectory tracking on a Franka Panda model under conservative partial actuation (torques applied only to joints 1–3). In nominal tracking, inverse-dynamics compensation provides clear benefit: CTC reduces joint-space RMSE by approximately 2.3× relative to PD (0.0052 m vs. 0.0121 m, aggregated over trajectories and speeds) while lowering torque RMS and saturation incidence. Under structured model mismatch and a sudden 0.5 kg payload step, ACTC is more robust than CTC, with markedly lower post-disturbance error (e.g., 0.0059 m vs. 0.0139 m post-payload RMSE) and substantially faster recovery (median ≈0.7–0.9 s across speeds, versus 3–> 5 s with speed-dependent censoring for CTC). Overall, the results support CTC when the model is accurate and ACTC when parametric uncertainty and recovery behavior dominate controller selection.

B. Future Work

Future work will extend the implementation to full 7-DOF torque control to quantify the benefit of commanding distal joints. Hardware validation will be performed under Franka Control Interface timing constraints (< 1 ms end-to-end). We will investigate learning-augmented adaptation to accelerate response to abrupt regime shifts while preserving stability characteristics.

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